



**BGSMath**  
BARCELONA GRADUATE  
SCHOOL OF MATHEMATICS

# Seminari Informal de Matemàtiques de Barcelona

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**Speaker:** Anna Jové.

**University:** Universitat de Barcelona.

**Date:** Wednesday, April 9th, 2025.

**Schedule:** 13:00 talk.

**Place:** UB (FMI aula T2) and Zoom.

**Language:** English.

**Title:** Universal covering maps. Different points of view.

**Abstract:** This talk is about universal covering maps for planar domains, that is, holomorphic maps from the unit disk to open domains of the plane, which are locally homeomorphic. Such functions (and its extension to the unit circle) are interesting from an analytic point of view, but also from the point of view of topology, geometry and measure; and can also be seen as an extension of Carathéodory's theory of prime ends for simply connected domains. I will discuss all this approaches and connections among them. Although the previous construction does not involve dynamics, some applications to dynamics will be presented. This is joint work with G. R. Ferreira.

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**About us:** *SIMBa* is a mathematics seminar organized by graduate students in the Barcelona area. It is aimed towards graduate and last-year undergraduate students. Our goals are disseminating knowledge from different branches of mathematics for those interested and promoting networking between the attendants.

This seminar is backed by the Faculty of Mathematics and Computer Science at Universitat de Barcelona, Faculty of Mathematics and Statistics at Universitat Politècnica de Catalunya, the Department of Mathematics at Universitat Autònoma de Barcelona, CRM, IMUB and BGSMath.

For more information, visit [seminari-simba.github.io/en](https://seminari-simba.github.io/en). You may contact us by sending an email to [seminari.simba@gmail.com](mailto:seminari.simba@gmail.com).